

Maximum Mark: 10

Quiz-2

Time: 55 minutes

1. (points) [0.5+3]
 - (a) Which orbital energy level is most affected by the Lamb-shift and what happens to its energy with respect to its fine structure levels.
 - (b) Draw the complete energy level representation for n=2 of a Hydrogen atom starting from Bohr's to Hyperfine structure with proper labels.
2. (points) [0.5+1+1+1] Define radial probability density function and find the properly normalized radial wavefunction of H-atom for ψ_{100} , ψ_{200} , ψ_{210} ? If there is any issue, please briefly comment on the same.
3. (points) [1+1+1] Compare the ratio of the frequencies of the absorption lines $1S \rightarrow 2P$ for the three isotopes 1H , 2D , and 3T of the Hydrogen atom by calculating the hyperfine shifts and splittings with the nuclear spin quantum numbers $I(H) = 1/2$, $I(D) = 1$, and $I(T) = 3/2$, given the nuclear magnetic moments $\mu_N(H) = 2.79\mu$, $\mu_N(D) = 0.857\mu$ and $\mu_N(T) = 2.98\mu$ where $\mu = e\hbar/2m_p$.

Useful Results

- $\psi_{100} = \frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0}\right)^{3/2} e^{(-r/a_0)}$;
- $\psi_{200} = \frac{1}{2\sqrt{2\pi}} \left(\frac{1}{a_0}\right)^{3/2} \left(1 - \frac{r}{2a_0}\right) e^{(-r/2a_0)}$;
- $\psi_{210} = \frac{1}{4\sqrt{2\pi}} \left(\frac{1}{a_0}\right)^{3/2} \left(\frac{r}{a_0}\right) e^{(-r/2a_0)} \cos(\theta)$;
- $E_{HFS} = C[F(F+1) - J(J+1) - I(I+1)]$ where C is a constant